AUTOMATED MEDICINE DISPENSER SYSTEM

A MINI PROJECT REPORT

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BONAFIDE CERTIFICATE

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DECLARATION

We affirm that the Mini Project Report titled **"AUTOMATED MEDICINE DISPENSER SYSTEM"** being submitted in partial fulfilment of the requirements for the award of Bachelor of Engineering is the original work carried out by us. It has not formed the part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

This project presents an automated tablet dispensing system using a microcontroller-based design. The system integrates an ultrasonic sensor, push buttons, LCD display, and stepper motors to provide a controlled and user-friendly medicine dispensing mechanism. The ultrasonic sensor detects the presence of a user, triggering the system to display instructions on the LCD. Users can select the required number of tablets for two different medicines using dedicated increment and decrement buttons. The selected quantities are displayed in real time on the LCD screen. Upon confirmation, stepper motors precisely rotate to dispense the specified number of tablets, ensuring accurate dosage control. Each motor rotates in fixed angular steps corresponding to individual tablet release. The system enhances safety, accuracy, and convenience in medicine dispensing, making it suitable for applications such as hospitals, pharmacies, and home healthcare. The use of automation reduces human error and ensures proper medication management.

SUSTAINABLE DEVELOPMENT GOAL (SDG):

The automated tablet dispensing system significantly contributes to improved healthcare by ensuring accurate and reliable medication management. One of the major challenges in healthcare, especially among elderly individuals and patients with chronic illnesses, is maintaining the correct dosage and timing of medicines. Manual handling often leads to errors such as missed doses, overdosing, or incorrect medication intake.

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CHAPTER 1
INTRODUCTION

In recent years, automation has played a vital role in improving healthcare systems by enhancing accuracy, efficiency, and reliability. One of the major challenges faced in hospitals, pharmacies, and home care environments is the proper management and dispensing of medicines. Errors in medication, such as incorrect dosage, missed intake, or confusion between different tablets, can lead to serious health complications, especially for elderly patients and individuals suffering from chronic illnesses.

To address these issues, an automated tablet dispensing system has been developed using embedded system technology. This system integrates components such as an ultrasonic sensor, microcontroller, LCD display, push buttons, and stepper motors to create a user-friendly and efficient medicine dispensing mechanism. The ultrasonic sensor detects the presence of a user and activates the system, ensuring energy efficiency and ease of operation.

The user can select the required number of tablets for different medicines using push buttons, and the selections are displayed on the LCD screen for clarity. Upon confirmation, the stepper motors rotate with precise angular control to dispense the exact number of tablets. This ensures accurate dosage and minimizes human intervention error and provide a reliable solution for safe and effective healthcare delivery. It is particularly useful in applications such as hospitals, pharmacies, and home-based patient care systems.

CHAPTER 2
LITERATURE REVIEW

2.1 REVIEW OF LITERATURE

The advancement of healthcare technology has led to the development of various automated systems aimed at improving medication management and patient safety. In traditional healthcare practices, medication dispensing is often performed manually, which increases the chances of human error, including incorrect dosage, missed medication schedules, and improper handling. To address these issues, researchers have explored different automated solutions over the years.

Initial developments in this field focused on simple electronic reminder systems that alerted patients to take their medicines at predefined times. Recent advancements in healthcare automation have led to the development of various automated tablet dispensing systems aimed at improving medication adherence and reducing human errors. Several studies have focused on designing efficient and user-friendly dispensing mechanisms. For instance, Divya et al. [1] presented a comprehensive survey on automatic pill dispensers, highlighting their importance in elderly care and reviewing different system designs. Similarly, Aung et al. [2] developed a pharmacy-based automated dispensing system that enhances efficiency and minimizes manual mistakes. Research has also emphasized the integration of advanced technologies such as human-machine interaction and robotics. Yuan et al. [3] explored collaborative systems that improve accuracy and reduce workload in healthcare environments.

Marmaglio et al. [4] compared different dispensing mechanisms and analyzed their performance in home healthcare applications. Furthermore, Olaleye et al. [5] conducted a bibliometric study that shows the increasing adoption of robotic tablet dispensing systems. Patient-centric improvements have also been a key focus. Amirthalingam et al. [6] studied patient satisfaction and found that automated systems significantly reduce waiting time and improve reliability. Ayimbila et al. [7] focused on personalized dosing, emphasizing the importance of precise drug delivery systems. In addition to automation, IoT-based solutions have gained attention.

Smith and Kumar [8] and Sharma et al. [9] developed smart medicine dispensers with remote monitoring capabilities, allowing better patient compliance. Embedded system-based designs have also been widely explored, as demonstrated by Nguyen and Lee [10] and Patel and Mehta [11], who implemented microcontroller-based dispensing systems for real-time operation. Automation in healthcare has further evolved with the use of intelligent and sensor-based systems.

Wang et al. [12] discussed the broader impact of automation technologies in drug dispensing, while Gupta and Singh [13] proposed a cost-effective Arduino-based solution. Kim and Park [14] introduced a sensor-based approach to enhance system accuracy and reliability. Recent studies have incorporated advanced technologies such as artificial intelligence and robotics. Chen et al. [15] developed an AI-based dispensing system to improve decision-making, while Reddy and Kumar. [16] highlighted the role of embedded systems in healthcare automation.

Ali et al. [17] introduced a smart pillbox with reminder features to assist patients in maintaining medication schedules. User interface and system usability have also been considered in modern designs. Nair and Das [18] developed a system with a simple interface to improve user interaction. Zhang et al. [19] explored robotic applications in healthcare, emphasizing automated drug dispensing systems.

Finally, Singh et al. [20] proposed an advanced IoT-based system with real-time monitoring and sensor integration.From the above studies, it is evident that automated tablet dispensing systems are continuously evolving with the integration of embedded systems, IoT, sensors, and artificial intelligence. However, there is still a need for cost-effective, simple, and reliable systems suitable for everyday use. The proposed system in this project aims to address these requirements by providing an efficient and user-friendly solution.

2.2 SUMMARY

Recent research highlights significant advancements in automated tablet dispensing systems aimed at improving medication adherence and reducing human errors. Various studies focus on embedded systems, IoT integration, and sensor-based technologies to enhance accuracy and efficiency. Automated systems with user-friendly interfaces and real-time monitoring have been developed for hospitals and home care. Advanced approaches include robotic dispensing and AI-based decision- making for precise drug delivery. Despite these improvements, many systems remain complex or costly. Therefore, there is a need for a simple, reliable, and cost-effective solution, which this project addresses by combining automation, ease of use, and accurate tablet dispensing.Advanced solutions also include robotic mechanisms and artificial intelligence for precise drug delivery and intelligent decision-making. Additionally, several designs focus on user-friendly interfaces, making them suitable for elderly patients and home healthcare applications. These systems have proven to reduce workload in hospitals and improve efficiency in pharmacies.

2.3 GAP ANALYSIS

- Existing systems are complex in design, making them difficult to implement and maintain. Many solutions use advanced technologies like IoT and AI, which increase the overall cost.
- Most systems are designed for hospitals or large-scale applications, not for home or personal use. Lack of simple and user-friendly interface, especially for elderly users.
- Some systems depend on internet connectivity, which reduces reliability in offline conditions. Limited focus on low-cost implementation, making them less accessible.
- Certain designs do not provide real-time response, leading to delays in dispensing. Some systems lack accuracy in dispensing mechanism, causing dosage errors.
- Maintenance and setup are complicated in many existing models. Power consumption is higher in advanced automated systems.

2.4 PROBLEM STATEMENT

In healthcare and home environments, proper medication management is essential for ensuring patient safety and effective treatment. However, traditional methods of storing and dispensing tablets are mostly manual, which increases the risk of human errors such as incorrect dosage, missed medication schedules, and confusion between different medicines. These issues are especially critical for elderly individuals, patients with chronic diseases, and those who require regular medication. Existing solutions like reminder alarms and basic pill organizers only assist in notifying users but do not ensure accurate dispensing of medicines. Advanced automated systems are available, but they are often expensive, complex, and require technical knowledge or continuous internet connectivity, making them unsuitable for widespread use. There is a need for a simple, reliable, and cost-effective system that can automatically dispense the correct number of tablets based on user input while minimizing human intervention. The system should be easy to operate, provide clear instructions, and ensure precise control over dosage. Therefore, the problem is to design and develop an automated tablet dispensing system that reduces medication errors, improves user convenience, ensures accurate dosage delivery, and is affordable and accessible for everyday use in healthcare and home settings.

CHAPTER 3

EXISTING METHOD

3.1 MANUAL COUNTING METHOD

The manual counting method is the most fundamental approach to tablet dispensing and is widely practiced in homes, small clinics, and pharmacies. In this method, a person physically counts tablets one by one and separates the required quantity based on a doctor's prescription. The process typically involves taking tablets from bulk containers or blister strips and placing them into another container or packet after counting. In some cases, tools such as counting trays or spatulas are used to make the process slightly more organized, but the responsibility of accuracy still lies entirely with the individual. This method is highly dependent on human concentration and precision. Even a small lapse in attention can result in errors such as undercounting or overcounting. For example, if a patient requires a strict dosage regimen, even a single tablet error can affect treatment outcomes. Human factors such as fatigue, repetitive strain, stress, and distractions significantly increase the likelihood of mistakes, especially when dealing with large quantities or multiple prescriptions at the same time. In busy pharmacy environments, the risk becomes even higher due to time pressure and workload.

Another important concern in manual counting is hygiene. Since tablets are handled directly, there is a risk of contamination from hands, surfaces, or environmental factors. Proper hygiene practices such as wearing gloves or using sanitized tools are not always strictly followed, particularly in non-clinical environments. This can compromise the safety and quality of medication. Additionally, the manual method lacks traceability and accountability. There is no record of how many tablets were dispensed or whether the correct dosage was given unless documented manually. This makes it difficult to verify or audit the process. The method is also inefficient in terms of time and labour, as it requires continuous human involvement. Despite its simplicity and low cost, the manual counting method is not suitable for situations where high accuracy, consistency, and safety are critical.

3.2 PILL ORGANIZER METHOD

The pill organizer method is designed to improve medication management by providing a structured way to store and access tablets. It typically consists of a container divided into multiple compartments, each labelled according to days of the week or times of the day (morning, afternoon, evening, night). The user or caregiver fills these compartments in advance based on the prescribed method helps reduce confusion, especially for individuals who take multiple medications daily. By visually organizing tablets, it allows users to quickly identify which medicines to take at a given time. It is particularly beneficial for elderly patients, people with memory issues, or those managing chronic conditions that require strict adherence to medication schedules.

However, the effectiveness of the pill organizer depends entirely on the accuracy of the initial setup. If tablets are placed incorrectly in the compartments, the error persists throughout the usage period, potentially leading to repeated incorrect dosages. The system does not have any built-in mechanism to detect or correct such mistakes. Furthermore, it does not provide any active reminders unless combined with external alarm systems. Another limitation is the lack of monitoring and feedback. The organizer cannot confirm whether the user has taken the medication or not. A missed dose may go unnoticed, and there is no alert system to notify caregivers. Additionally, the method requires regular refilling, which can be inconvenient and time- consuming. Hygiene can also be a concern if the compartments are not cleaned properly between uses. Despite these limitations, the pill organizer remains a popular solution due to its simplicity, affordability, and ease of use. However, it does not eliminate human dependency and does not offer automation or intelligent control.

3.3 ALARM BASED REMINDER SYSTEM

Alarm-based reminder systems are designed to assist users in maintaining a consistent medication schedule by providing timely alerts. These systems use electronic timers, buzzers, or mobile notifications to remind patients when it is time to take their medication. They can be implemented using simple devices such as alarm clocks or advanced platforms like smartphone applications. The primary advantage of this method is its ability to improve adherence to prescribed schedules. By providing regular reminders, it reduces the likelihood of missed doses, especially for individuals with busy lifestyles or memory-related challenges. Some advanced systems allow customization of multiple alarms for different medications, making them suitable for complex treatment plans.

However, alarm-based systems do not address the issue of dosage accuracy. While they remind the user to take medication, they do not control or verify the number of tablets consumed. The user must still manually select and take the correct tablets, which introduces the possibility of human error.

Additionally, there is no guarantee that the user will respond to the alarm. Alerts can be ignored, silenced, or forgotten, particularly if the user is engaged in other activities or is not near the device.
Another limitation is the lack of confirmation and monitoring. The system cannot track whether the medication was actually taken after the alarm was triggered. This makes it difficult for caregivers or healthcare providers to ensure compliance. Some advanced reminder applications attempt to address this by requiring user input or acknowledgment, but these features still rely on user honesty and participation.

Furthermore, alarm-based systems may become less effective over time due to habituation, where users start ignoring frequent alerts. Technical issues such as battery failure, incorrect time settings, or software glitches can also affect reliability. While this method is cost-effective and easy to implement, it does not provide a complete solution for medication management, as it lacks automation, verification, and dispensing capabilities.

3.4 LIMITATIONS

Despite the usefulness of existing methodologies such as manual counting, pill organizers, and alarm- based reminder systems, several limitations reduce their effectiveness in ensuring safe and accurate medication management. One of the major limitations is the high dependency on human involvement. In methods like manual counting and pill organizers, the accuracy of dosage completely relies on the user or caregiver. Any mistake in counting or arranging tablets can lead to incorrect medication intake, which may cause serious health risks. Another important limitation is the lack of accuracy and precision. Manual methods are prone to errors such as overcounting or undercounting, especially when handling large quantities or multiple medications. Similarly, pill organizers do not verify whether the correct tablets are placed in each compartment, leading to possible repeated errors. The absence of automation is also a significant drawback. Most existing methods do not include automatic dispensing mechanisms, requiring users to manually take medicines even after reminders. This increases the chances of missed doses or incorrect intake.

CHAPTER 4

PROPOSED METHOD

4.1 INTRODUCTION

The proposed automated tablet dispensing system is developed to provide a reliable and efficient solution for medication management. In traditional methods, dispensing tablets manually can lead to errors such as incorrect dosage, missed medication, and confusion between different medicines. These issues can have serious consequences, especially for elderly patients and individuals with chronic diseases who depend on timely and accurate medication. To overcome these challenges, the system integrates modern embedded technology with simple user interaction. It combines components such as a microcontroller, ultrasonic sensor, LCD display, push buttons, and stepper motors to automate the dispensing process. The design focuses on accuracy, ease of use, and cost-effectiveness, making it suitable for both healthcare environments and home use. The system ensures that users can easily select the required number of tablets and receive the exact dosage without manual counting. By reducing human intervention and providing clear instructions, the system enhances safety and improves medication adherence.

4.2 WORKING PRINCIPLE

The working principle of the automated tablet dispensing system is based on the coordination between sensing, user input, and controlled mechanical operation. Initially, the ultrasonic sensor continuously monitors the surroundings to detect the presence of a user. When a person is detected within a predefined distance, the system is activated. The LCD display then shows a message prompting the user to enter the number of tablets required. The user interacts with the system using push buttons to increase or decrease the count of tablets for different medicines. The selected values are displayed on the LCD in real time, allowing the user to verify the input. Once the desired quantity is set, the user presses the confirmation button to proceed. After confirmation, the microcontroller processes the input and calculates the number of steps required for the stepper motors. Each stepper motor is connected to a specific medicine container and rotates by a fixed angle corresponding to the release of one tablet. For example, a fixed angular rotation (such as 45°) ensures that exactly one tablet is dispensed per step cycle.

The stepper motors are driven precisely to rotate the required number of steps based on the selected count. This controlled motion ensures accurate dispensing without overlap or error. Once all tablets are dispensed, the system displays a completion message on the LCD. Finally, the system resets the count and returns to standby mode, waiting for the next user interaction. This working principle ensures accurate, efficient, and automated tablet dispensing with minimal human intervention.

4.2.1 BLOCK DIAGRAM

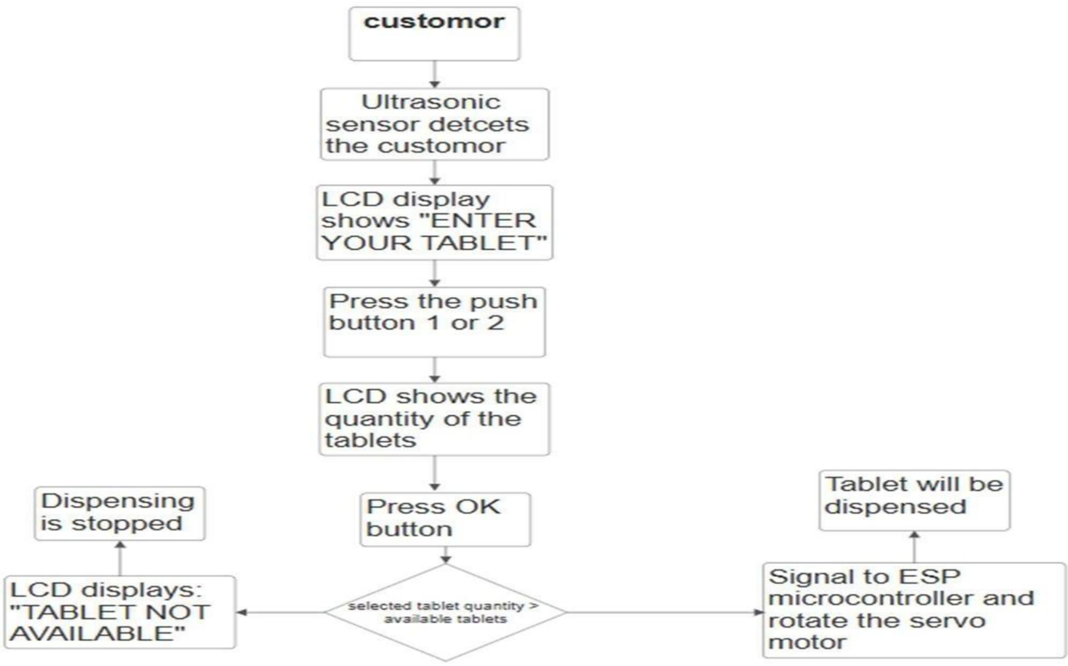

Figure 4.1 Working of the Automated medicine dispenser system

The Figure 4.1 illustrates the working flow of an IoT Based Room Status Monitoring and Indicating System using an ESP32 microcontroller. The system is equipped with current sensors connected to electrical loads such as lights and fans to continuously monitor their operational status. When an appliance is switched ON, the current sensor detects the flow of current and sends the signal to the ESP32. The microcontroller processes this input and identifies the condition as an active state. Similarly, when the appliance is switched OFF, the absence of current is detected, and the system updates the status as inactive. These signals are continuously processed by the ESP32, which updates the room status in real time based on the input received from the sensors.

The updated status is then displayed through a web based dashboard for monitoring purposes as shown in Figure 4.1. The dashboard provides real time information about the ON/OFF condition of electrical appliances, allowing users to monitor the system remotely through web or mobile interfaces.

4.2.2 ESP32 DEV MODULE

The Figure 4.2 illustrates the ESP32 Microcontroller, an advanced and widely used platform for developing IoT based electronic systems. It consists of integrated hardware with built in Wi Fi and Bluetooth capabilities, along with software support through the Arduino IDE for programming. The ESP32 is commonly chosen due to its high processing speed, flexibility, and low cost, making it suitable for real time applications. As shown in Figure 4.2, the board receives input signals from current sensors and processes the data to determine the status of electrical appliances. It also controls output devices such as relay modules and communicates with a web based dashboard for real time monitoring. In this project, the ESP32 acts as the main controller that processes sensor data, identifies ON/OFF conditions of loads, and updates the system status accordingly. Its reliability, connectivity, and adaptability make it an ideal choice for IoT based energy monitoring applications, reinforcing the system architecture depicted in Figure 4.2.

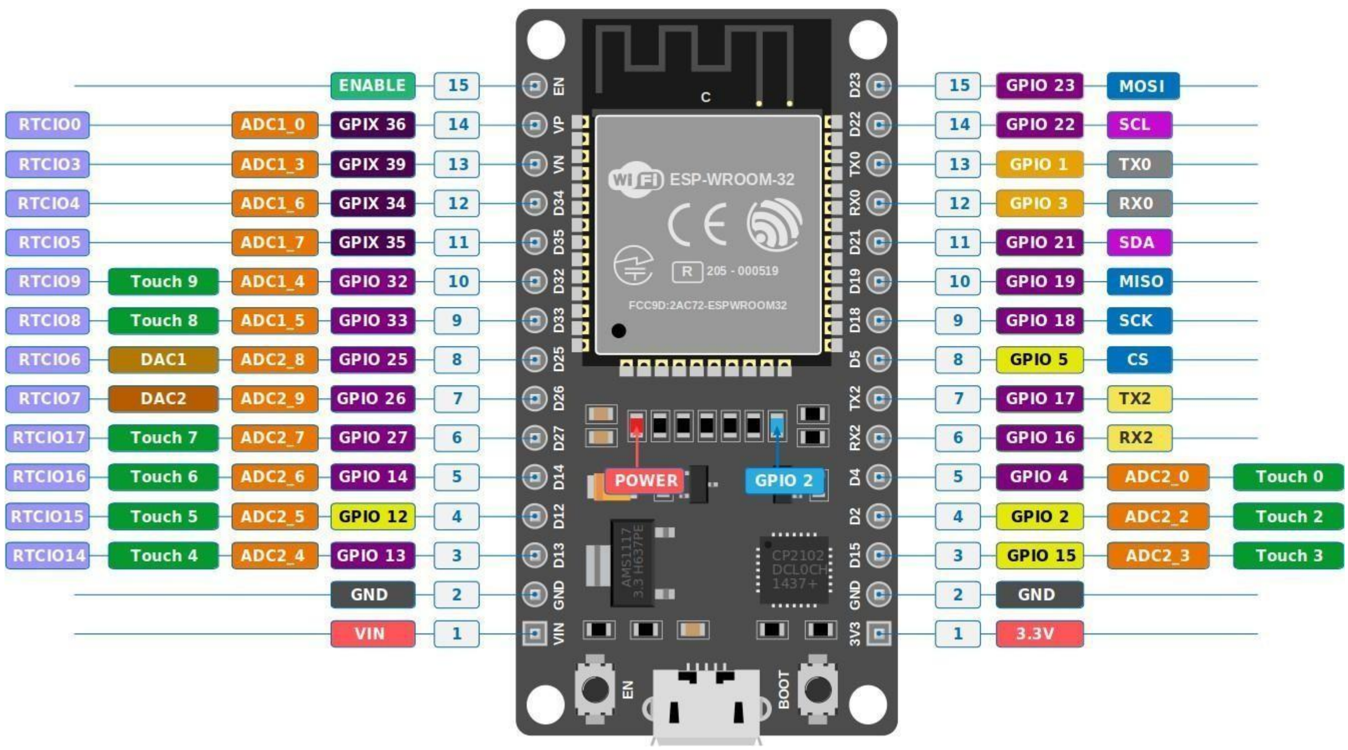

Figure 4.2 ESP32 Microcontroller

The Figure 4.2 explains the ESP32 microcontroller, showing its role as the main controller that reads current sensor inputs and manages outputs such as relay modules and web based dashboard communication.

4.2.3 ULTRA SONIC SENSOR

HC-SR04 Ultrasonic Sensor to detect the presence of a user and control when the system should start accepting input. This sensor works by emitting ultrasonic sound waves and measuring the time it takes for the echo to return after hitting an object (like a hand). It has two main pins: TRIG and ECHO. In your code, the TRIG pin sends a short 10-microsecond pulse, which causes the sensor to release ultrasonic waves. When these waves bounce back, the ECHO pin stays HIGH for a duration proportional to the distance of the object.

Figure 4.3 Ultra Sonic sensor

Figure 4.3 illustrates the Ultra sonic sensor module, showing its structure and pin configuration used for measuring electrical current. It highlights how the sensor detects current flow and provides an analog output for processing by the microcontroller.

4.1.1 STEPPER MOTOR DRIVER

The motor driver acts as the interface between the microcontroller and the stepper motors, ensuring that the motors receive the correct voltage, current, and switching sequence required for precise movement. Since a microcontroller cannot directly drive a stepper motor due to its limited current output, the driver amplifies the control signals and energizes the motor coils in the proper order. For commonly used motors like the 28BYJ-48, a driver module such as the ULN2003 Motor Driver Board is typically used. It takes low-power signals from the controller and converts them into higher-power outputs that rotate the motor step by step. In this tablet dispensing system, the motor driver enables accurate 45° rotations for each count, allowing tablets to be dispensed in controlled quantities while protecting the microcontroller from electrical overload.in Figure 4.4.

Figure 4.4 MOTOR DRIVER

Figure 4.4 explains the Blynk IoT application, showing how it connects the ESP8266 MOTOR DRIVER.

4.2.3 ESP 32 STUMILATION

ESP32 is a complete, low-code IoT software platform designed to help users prototype, deploy, and remotely manage connected electronic devices at any scale. It allows developers and businesses to connect hardware to the cloud and build iOS, Android, and web applications with drag- and-drop interfaces, enabling real-time device control and automation, as shown in Figure 4.5. The platform provides flexibility for monitoring and managing devices remotely, making it suitable for both small projects and large-scale industrial applications, shown in Figure 4.5.

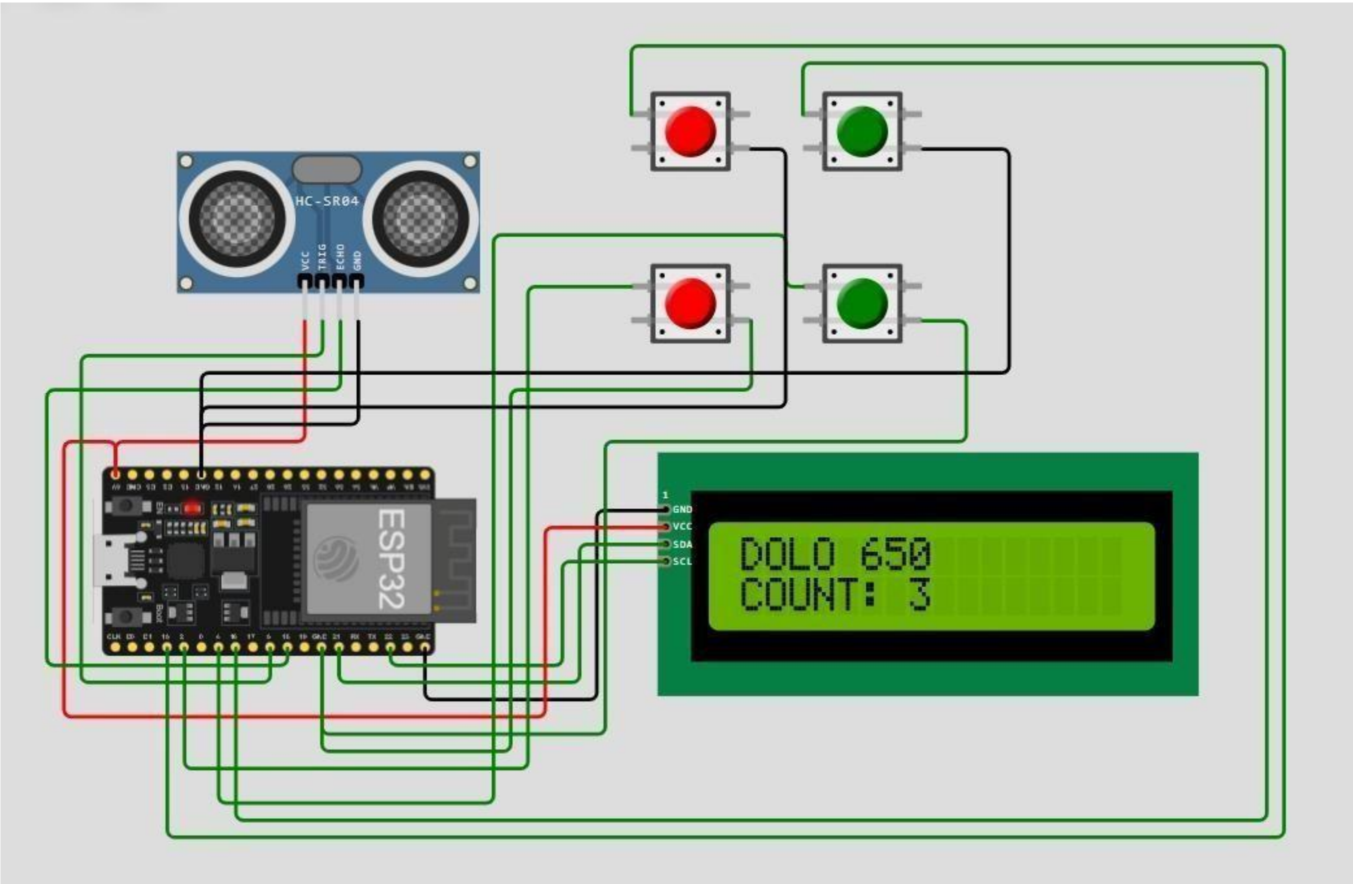

Figure 4.5 Blynk Web Dashboard

Figure 4.5 illustrates the WOKKI platform, showing how hardware devices like the ESP32 Node MCU connect to the cloud and transmit real-time data. It highlights the dashboard interface where inputs, outputs, and detected persons are monitored, enabling remote control and smart automation.

4.1 WORKING OVERVIEW

This project works as an automated, touchless tablet dispensing system that activates only when a user is detected. An HC-SR04 Ultrasonic Sensor continuously measures distance, and when a hand is placed within about 10 cm, the system turns on and prompts the user via the LCD to enter the number of tablets. The user selects the required quantity for each medicine (Dolo and Brufen) using push buttons, with the values shown on the display for confirmation. Once the confirm button is pressed, the system processes the input and drives the stepper motors using a motor driver like the ULN2003 Motor Driver Board. Each motor rotates in fixed steps (45° per count), dispensing the exact number of tablets. After completing the operation, the display shows a “DONE” message, resets the counts, and the system returns to a waiting state until the next user is detected.

4.3 ADVANTAGES

- Uses the HC-SR04 Ultrasonic Sensor to detect a hand, avoiding physical touch and improving hygiene
- Stepper motors rotate in fixed angles (45° per count), ensuring precise quantity delivery.
- A driver like the ULN2003 Motor Driver Board protects the microcontroller and provides sufficient power to the motors.
- Push buttons make it easy to increase/decrease tablet count.
- LCD shows instructions, selected counts, and system status (WAITING, DISPENSING, DONE).
- Can handle more than one type of tablet (e.g., Dolo and Brufen)
- Automated counting minimizes mistakes in manual dispensing.

4.4 PROJECT SUMMARY

The automated tablet dispensing system is a practical and efficient solution designed to address the limitations of traditional medication management methods. By integrating a microcontroller, ultrasonic sensor, LCD display, push buttons, and stepper motors, the system provides accurate and automated tablet dispensing with minimal human intervention. The project focuses on improving healthcare safety by ensuring correct dosage and reducing the chances of errors. It enhances user convenience through a simple interface and reliable operation. The system is particularly beneficial for elderly individuals, patients with chronic conditions, and healthcare providers who require consistent and precise medication handling. Overall, the project demonstrates how embedded system technology can be effectively applied to solve real-world healthcare challenges, offering a balance between accuracy, affordability, and ease of use.

CHAPTER 5
RESULTAND DISCUSSION

5.1 PRACTICAL RESULTS

The automated tablet dispensing system was successfully implemented and tested under various operating conditions to evaluate its performance and reliability. The system was able to detect user presence accurately using the ultrasonic sensor and activate only when required, demonstrating efficient operation and reduced power usage. During testing, the push button interface allowed users to easily select and adjust the number of tablets for each medicine. The LCD display provided clear and real-time feedback, ensuring that the user could verify the selected values before confirmation. This improved user interaction and minimized input errors. The stepper motors performed with high precision, dispensing the exact number of tablets corresponding to the selected count. For each unit count, the motor rotated by a fixed angle, ensuring consistent and accurate tablet release. Multiple trials confirmed that the system maintained consistent performance without variation in output. The response time of the system was observed to be satisfactory, with quick detection, input processing, and dispensing actions. The overall system operation was smooth, and no major delays or malfunctions were encountered during testing. Additionally, the system demonstrated reliable performance over repeated cycles, indicating good durability and stability. The automatic reset after dispensing ensured that the system was ready for the next operation without manual intervention.

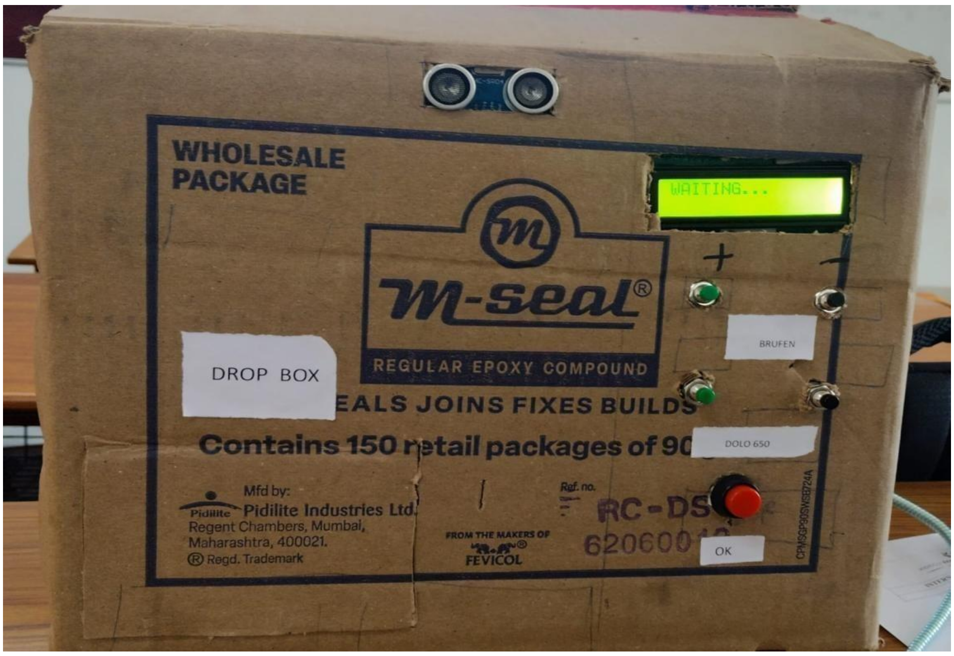

Figure 5.1 FRONT VIEW OF THE DISPENSER

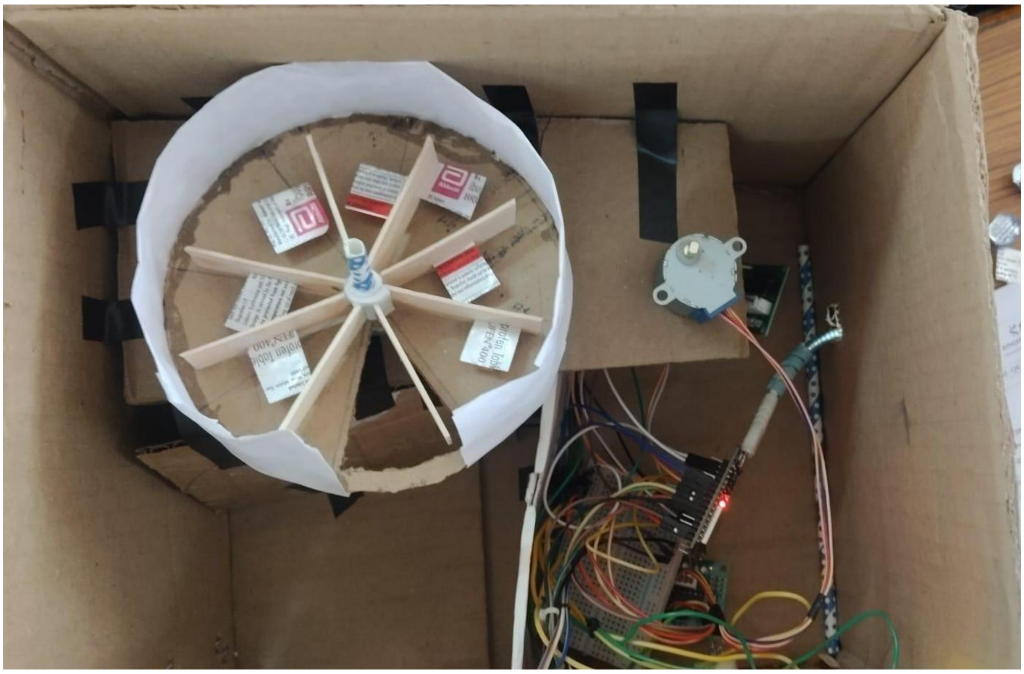

Figure 5.2 INNER VIEW OF THE DISPENSER

5.2 IMPACT AND INNOVATION

The automated tablet dispensing system significantly improves medication management by ensuring accurate dispensing and reducing human errors such as incorrect dosage and missed medication. It enhances patient safety and treatment effectiveness, especially for elderly individuals and patients with chronic illnesses who require regular medication. The system reduces the workload of caregivers and healthcare professionals by automating the dispensing process, saving time and ensuring consistent performance. It also promotes better hygiene by minimizing direct handling of tablets and is suitable for use in homes, hospitals, and pharmacies. Being cost-effective and easy to operate, it is accessible to a wide range of users and supports improved healthcare management.

The innovation of this project lies in its ability to combine simplicity with effective automation. The use of an ultrasonic sensor enables automatic user detection and activates the system only when needed, improving energy efficiency. Stepper motors are utilized for precise control, ensuring accurate tablet dispensing without the need for complex mechanisms. The integration of an LCD display and push-button interface provides a user-friendly experience with clear instructions and easy input control. The system does not require internet connectivity, making it reliable and practical for everyday use. Overall, the design focuses on affordability, accuracy, and ease of use, making it a highly effective solution for real-world healthcare applications.

CHAPTER 6

CONCLUSION

The automated tablet dispensing system provides an effective solution to the challenges associated with traditional medication management methods. By integrating a microcontroller with sensors, user interface components, and precise motor control, the system ensures accurate and reliable dispensing of tablets. It successfully reduces human errors such as incorrect dosage, missed medication, and confusion between different medicines The system offers a user-friendly interface that allows easy interaction, making it suitable for people of all age groups, especially elderly patients and individuals with chronic conditions. The use of stepper motors ensures precise control over tablet dispensing, while the ultrasonic sensor enhances efficiency by activating the system only when needed. In addition, the design is cost-effective, compact, and does not require complex infrastructure or internet connectivity, making it practical for use in homes, hospitals, and pharmacies. The consistent performance and ease of operation make the system a dependable tool for improving medication adherence and healthcare management.

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